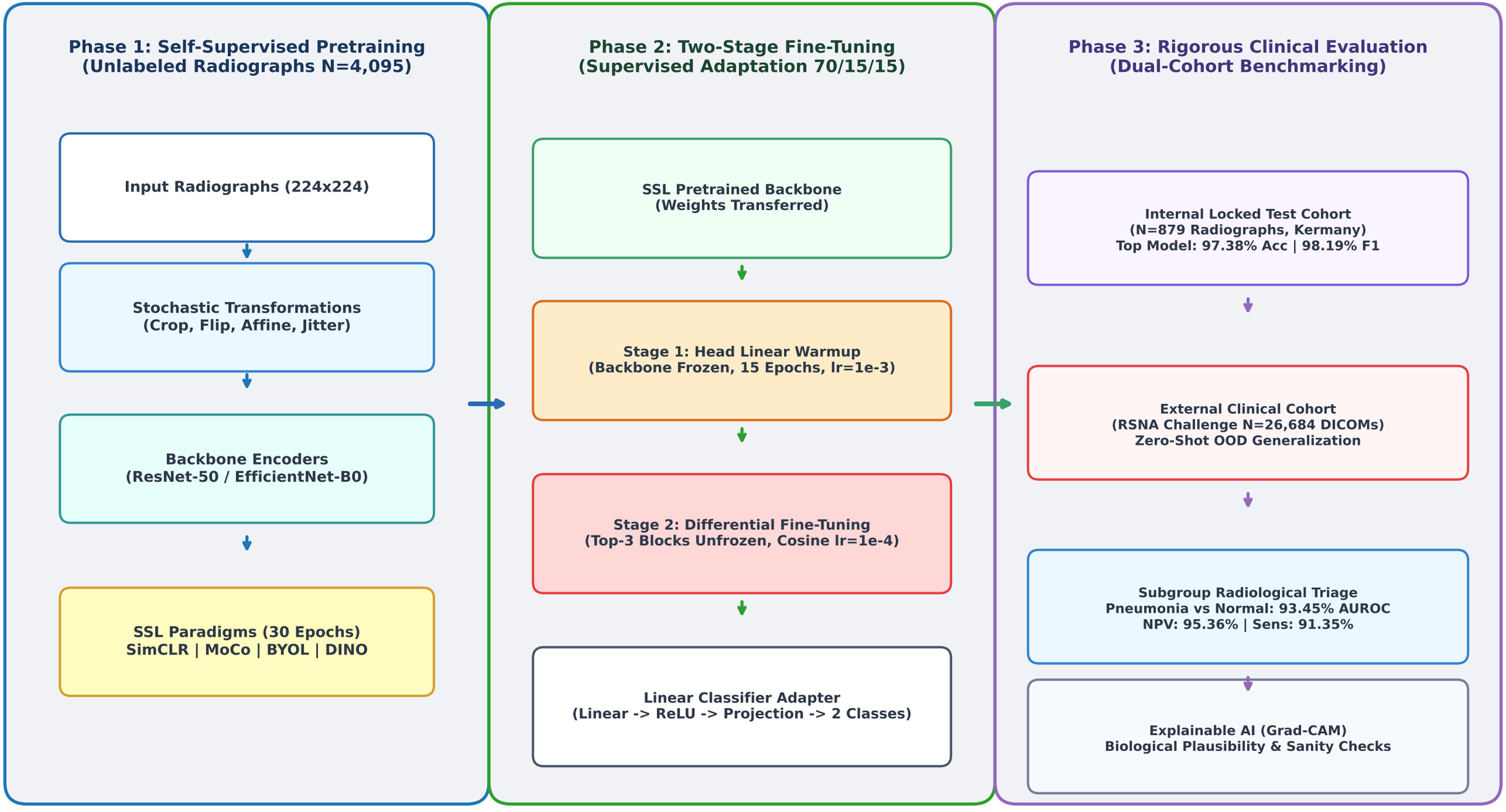
Phase 1: Self-Supervised Pretraining
(Unlabeled Radiographs N=4,095)
Input Radiographs (224x224)
Stochastic Transformations
(Crop, Flip, Affine, Jitter)
Backbone Encoders
(ResNet-50 / EfficientNet-B0)
SSL Paradigms (30 Epochs)
SimCLR | MoCo | BYOL | DINO
Phase 2: Two-Stage Fine-Tuning
(Supervised Adaptation 70/15/15)
SSL Pretrained Backbone
(Weights Transferred)
Stage 1: Head Linear Warmup
(Backbone Frozen, 15 Epochs, lr=1e-3)
Stage 2: Differential Fine-Tuning
(Top-3 Blocks Unfrozen, Cosine lr=1e-4)
Linear Classifier Adapter
(Linear -> ReLU -> Projection -> 2 Classes)
Phase 3: Rigorous Clinical Evaluation
(Dual-Cohort Benchmarking)
Internal Locked Test Cohort
(N=879 Radiographs, Kermany)
Top Model: 97.38% Acc | 98.19% F1
External Clinical Cohort
(RSNA Challenge N=26,684 DICOMs)
Zero-Shot OOD Generalization
Subgroup Radiological Triage
Pneumonia vs Normal: 93.45% AUROC
NPV: 95.36% | Sens: 91.35%
Explainable AI (Grad-CAM)
Biological Plausibility & Sanity Checks